

Brady Buten

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Robotics controls engineer building autonomy on real hardware: UKF, MPC, and A* on a field-tested deterrent robot; embedded vehicle controls, CAN interlocks, and HIL benches at Equipment Technologies; PLC and vision integration at GM. Full-time starting May 2027

Education

University of Michigan – Ann Arbor, MI

M.S.E., Electrical & Computer Engineering (Control Systems Concentration)

May 2027

- Robotics Lab (ROS2, SLAM), Online Learning for Multi-Agent Control, Self-Driving Cars, Model Predictive Control

B.S.E., Mechanical Engineering (Robotics Concentration) & B.S.E., Computer Engineering – GPA: 3.93 / 4.00

May 2026

- Summa Cum Laude, James B. Angell Scholar – 6 consecutive terms of straight A's across dual engineering degrees
- Delivered lectures and mentored 350+ students across 4 semesters as Lead Thermodynamics Teaching Assistant

Work Experience

Equipment Technologies (self-propelled agricultural sprayers) – Mooresville, IN

Vehicle Controls & Software Engineering Intern

May 2026 – Aug 2026

- Engineered end-to-end engine cooling fan control, reducing blade pitch oscillations by 76% with 15+ hours of field validation
- Designed Ag Leader autosteer disengage on a 2° threshold and 300ms debounce, preventing steering lockout from rapid input
- Programmed a fault-tolerant park brake interlock on 2 CAN and 1 hardwired input, escalating faults and lockout below 1.5 mph
- Built HIL benches closing sensor-controller-actuator loops across 5 subsystems, automating fault injection and signal logging

General Motors: Factory ZERO – Detroit, MI

Controls Engineer Intern – Paint Shop

May 2025 – Aug 2025

- Analyzed 8 weeks of FANUC plug insert robot faults, driving teach position optimizations to cut daily faults from 400+ to ~25
- Integrated OCR vision with Allen-Bradley PLCs over EtherNet/IP to flag charge-port mismatches, cutting 20+ hrs/week downtime
- Diagnosed windshield error-proofing vision, correcting thresholds to cut faults by 87% and prevent \$15K+ in monthly scrap

Projects

Michigan DNR Sponsored Project: Autonomous Turkey Deterrent Robot – Ann Arbor, MI

Technical Lead – Controls & Vision

Jan 2026 – Apr 2026

Presented to ME External Advising Board · Selected for Departmental Video Release · Best Front-End Design Award

- Directed a 4-person controls & vision subteam for a waypoint-driven autonomous robot dog, deterring live turkeys in field trials
- Implemented a UKF fusing 100 Hz IMU and 1 Hz GPS, validated in a custom sim modeling dynamics and mapped obstacles
- Integrated a YOLO model feeding live detections into an occupancy costmap, driving A* path planning at 2 Hz across 80x40 m
- Built MPC on Raspberry Pi for trajectory tracking and obstacle avoidance, enforcing speed constraints with <40ms solve times
- Designed a behavior controller sequencing 9 predator display modes across servos, audio, and base motion at <0.3s latency

Sartorius Sponsored Project: Custom Live Cell Incubator – Ann Arbor, MI

Project Lead – Controls & Software

Jan 2025 – Dec 2025

- Led a 6-person cell-incubator build, integrating across mechanical, embedded, and controls to reach 37°C ±0.5 and >90% RH
- Constructed a prototype testbed to characterize thermal/humidity actuators, driving a redesign that cut response times by 62%
- Developed the RPi control stack from scratch and a custom STM32 serial protocol, holding setpoints under enclosure leaks
- Recovered 3 weeks of slip on a \$13K budget, 9-vendor build through structured check-ins, closing a week ahead of schedule

Motion Paint: Gesture Controlled Memory Game – Ann Arbor, MI

Embedded Systems Developer

Oct 2024 – Dec 2024

- Wrote embedded C firmware across two STM32 boards, including drivers for a custom 150-LED grid and a 2x16 LCD screen
- Designed 4-corner IR-camera calibration mapping operator pose to 2D coordinates, correcting for unknown mounting rotation
- Architected a 31-state FSM with custom board-to-board protocols, driving gesture sensing, haptic feedback, and audio triggers

Technical Proficiencies

- Controls & Estimation:** MPC, LQR, PID, Optimal Control, UKF/EKF, Sensor Fusion, State Estimation, System Identification
- Planning & Autonomy:** Trajectory Optimization, Motion Planning, A*, OMPL, Behavior Trees, ROS2, Nav2, MoveIt2, SLAM
- Embedded & Software:** C/C++, Python, MATLAB/Simulink, EtherNet/IP, I2C/UART, CAN, HIL/SIL, Gazebo, Linux, Git